Title: Bayesian neural network with autoencoder for model-based description of α-particle preformation factor
Manuscript ID: CN10966

Response to the reviewer's comments and suggestions

We sincerely appreciate your constructive comments and suggestions, which are useful and have helped us to improve the quality of our work significantly.

I. Overall Assessment and Recommendation

This manuscript presents a hybrid machine learning framework combining Bayesian neural networks with autoencoders (BNN-Auto) to describe the α-particle preformation factor P_α within the cosh potential (CPT) model for α decay. The topic is timely and the general approach has merit, combining modern probabilistic machine learning techniques with a well-established nuclear decay model. The reported improvements in RMS deviations over the traditional least-squares approach are noteworthy, and the extension to predict half-lives for $Z = 120$ superheavy nuclei is interesting.

Response: We sincerely thank you for your meticulous review of our work and for your recognition and positive evaluation of our work.

II. Major Scientific and Methodological Concerns

1. Inconsistency in feature description (Section III, p. 8): The authors state that "Input features comprise physical quantities such as proton number N, neutron number Z...", but this reverses the standard notation. Throughout nuclear physics, Z is the proton number and N is the neutron number. In Section II.B, the correct assignment is used. This inconsistency is a serious error that must be corrected. The authors must carefully check the entire manuscript for consistency in this notation.

Response: Thank you for your meticulous review of the manuscript and for pointing out this crucial error. In Section III (page 8), we mistakenly reversed the symbols for proton number and neutron number. This is indeed a serious typing mistake, and we sincerely apologize for it. We have corrected the original text to "proton number Z, neutron number N" as the standard format, as per your suggestion. Additionally, we have thoroughly checked the entire document and standardized the use of symbols throughout to ensure consistency in the terminology used in nuclear physics.

2. Lack of comparison with other ML methods: The paper compares BNN-Auto only against the CPT LSM method. For a paper that positions itself as introducing a novel ML framework to the field, it is essential to compare against other machine learning approaches used for similar nuclear physics predictions, e.g., standard feedforward neural networks, random forests, Gaussian processes, or at least a standard BNN without the autoencoder component. Without such comparisons, the reader cannot assess the specific added value of the autoencoder architecture versus the Bayesian framework versus the combination.

Response: Thank you for your important suggestions. We fully agree that comparing the newly proposed machine learning framework with the commonly used methods in the field is crucial for evaluating its added value. Therefore, we have systematically supplemented the revised manuscript and compared BNN-Auto with several mainstream machine learning methods that you mentioned on the same nuclide dataset. As shown in Figure 5 in the revision, in the prediction of the logarithm of the preformation probability α decay ($log_{10}P_\alpha$), the proposed BNN-Auto framework achieved the lowest root mean square error ($\sigma_{RMS} = 0.346$), which was significantly better than random forest (RF, 0.351), Gaussian process (GP, 0.384), standard feedforward neural network (SFNN, 0.392), and the simple baseline linear least squares regression (L-LSR, 0.847). This comparison established the superiority of BNN-Auto in this study in terms of prediction accuracy. More importantly, compared with these deterministic models, the endogenous Bayesian framework within BNN-Auto enables it to simultaneously provide calibrated predictions of uncertainty, which is a key advantage for its reliable extrapolation predictions in data-scarce regions such as superheavy nuclei. This comparative analysis and Figure 5 have been included in the "Results and Discussion" section of the revised manuscript.

3. Insufficient justification of the autoencoder architecture: The paper claims the autoencoder "enhances the robustness of feature representation," but this claim is never substantiated with an ablation study. What is the performance of BNN alone (without the autoencoder)? What is the performance of a standard neural network with autoencoder but without Bayesian inference? Without these comparisons, the contribution of each component remains unclear.

Response: Thank you for pointing this out. We fully agree that it is essential to quantitatively evaluate the contribution of each component in the framework through ablation studies. Following your suggestion, we designed and conducted rigorous ablation experiments, and the results are also summarized in Figure 5 of the revised manuscript. By constructing a standard Bayesian neural network (S-BNN) without an autoencoder, we obtained a prediction error of $\sigma_{RMS} = 0.386$, and it was significantly higher than the complete BNN-Auto framework. This directly confirmed that the autoencoder component is not redundant. It learns the latent feature representations through reconstruction loss, which can effectively extract more robust and essential physical correlations from the input kernel features, thereby substantially improving the generalization ability of the prediction model. Then, we removed the KL divergence regularization term in the loss function, that is, turned off Bayesian inference, and denoted it as KL off. We only retained the reconstruction loss of the autoencoder and the MSE loss of the prediction. The performance of this model also showed a significant decline. These results indicate that the superiority of the BNN-Auto framework stems from the synergy between the autoencoder (responsible for learning robust feature representations) and the Bayesian framework (responsible for providing regularization and uncertainty quantification). We have elaborated on these results in the "Results and Discussion" section of the revised manuscript.

4. The weighting factor k in Eq. (10): The loss function in Eq. (10) introduces a weighting factor k that balances reconstruction and prediction losses. The chosen value of k is never reported. This is a critical hyperparameter, and its value, sensitivity analysis, and

optimization procedure should be clearly documented. How does the model performance change with different values of k?

Response: Thank you for your attention to this crucial hyperparameter. You are absolutely correct. The value of the weight factor k, the optimization process, and their impact must be clearly stated to ensure the reproducibility of the research. We have provided additional explanations in the "Results and Discussion" section of the revised manuscript: Through systematic grid search and validation set performance evaluation, we chose $k = 0.2$. To fully demonstrate its influence and verify the rationality of the selection, we conducted a detailed sensitivity analysis, and the results are presented in Figure 6 of the revised manuscript. As shown in Figure 6, we scanned the values of k from 0 to 1. In the complete BNN-Auto framework (KL on), the validation set error reaches its minimum at $k = 0.2$, and remains low and stable within a wider range of $k \in [0.1, 0.6]$. This proves the robustness of our selection. Further comparison of the "KL on" curve with the KL off curve (which removes KL regularization) reveals that in the interval $k \in [0.4, 0.6]$, the performance of KL on is significantly better, highlighting the crucial role of Bayesian regularization in preventing performance degradation during weight reconstruction. When $k > 0.6$, the two curves merge and eventually deteriorate sharply as k approaches 1, which indicates that when the loss function is overly biased towards the reconstruction task, the prediction target is ignored, and any regularization cannot remedy the situation, thereby confirming the rationality of our choice of $k = 0.2$. Your suggestion ensures the transparency of the hyperparameter selection and deepens the understanding of the model behavior.

5. Overfitting concerns and validation strategy: While the authors perform 10-fold cross-validation repeated 5 times, the results in Table I show that for "all nuclei," the training set improvement (30.75%) is lower than the validation set improvement (35.37%), while for even-even nuclei the training improvement (61.14%) exceeds the validation improvement (54.49%). This inconsistent pattern across subsets is not discussed and warrants explanation. Furthermore, with 535 nuclei and the described network architecture, the ratio of parameters to data points should be explicitly stated to assess potential overfitting.

Response: Thank you for raising these important points regarding the evaluation of our model. We appreciate the opportunity to clarify. The observed pattern where the validation improvement slightly exceeds the training improvement for the full, heterogeneous dataset while the opposite holds for the homogeneous even-even subset can be attributed to the intrinsic regularization of our Bayesian framework. The variational inference prevents overfitting to the complex, mixed-nuclei training set, leading to a conservative yet more generalizable model, which is reflected in the strong validation performance. Conversely, the model can optimize more thoroughly on the physically consistent even-even nuclei, showing a more conventional pattern. Regarding the concern about overfitting, we have explicitly detailed the data split in the revised manuscript, confirming that in each fold, 481 nuclei are used for training. Our BNN-Auto model contains approximately 4.4×10^4 parameters, resulting in a parameters to training data ratio of about 1:82.5, which is substantially less than 1. This, combined with the built-in regularization from the KL divergence and the autoencoder's reconstruction loss, effectively mitigates overfitting risk and ensures robust generalization, as evidenced by the consistent performance gains on all validation sets. We

have incorporated these clarifications and the explicit parameter count into the manuscript.

6. Uncertainty quantification is underexplored: One of the main advertised advantages of BNN is uncertainty quantification. However, the uncertainty estimates are only briefly mentioned (20 stochastic forward passes) and are not systematically analyzed. Are the predicted uncertainties well calibrated? Do they cover the true experimental values at the expected confidence level? A calibration analysis (e.g., coverage probability plot) would substantially strengthen the paper.

Response: Thank you for this crucial and insightful suggestion. We fully agree with your viewpoint on the importance of systematically analyzing the calibration of the uncertainty in Bayesian Neural Network (BNN) predictions. As you pointed out, uncertainty quantification is one of the core advantages of BNN, and a rigorous assessment of its reliability is crucial for the credibility of the results. First of all, please allow us to explain that in the current revision, we have not included such systematic and global calibration analysis charts like coverage probability graphs. We understand that this is the most direct and ideal way to fully verify uncertainty quantification, and we will conduct in-depth exploration and supplementation in subsequent work. However, we hope to provide a reasonable explanation and response to your questions regarding "whether the prediction uncertainty is well calibrated" and "whether it covers the true value at the expected confidence level" based on the existing verifiable evidence in the paper. Your concern about the sampling frequency $T = 20$ (Comment 11) is directly related to the reliability of the uncertainty estimation. As we demonstrated in the detailed reply to Comment 11 (i.e., the new Figure 7 added in the revised manuscript), through empirical convergence analysis, when $T \geq 15$, the predicted mean and standard deviation (i.e., the quantification of uncertainty) have been basically stable. This means that the uncertainty estimation obtained using $T = 20$ sampling is statistically convergent and numerically stable, and it does not result in drastic fluctuations due to insufficient sampling. This is a necessary prerequisite for evaluating its calibration. The core results in the paper (Figure 2, Figure 4, Table I) provide indirect but powerful support for the rationality of the uncertainty estimation. In Figure 2, we take nuclides such as Pu, Cm, Am, At as examples, and intuitively show that the corrected prediction (dark blue posterior band) is not only closer to the experimental value (green line), but also the prediction uncertainty interval is significantly narrowed and always includes the experimental value. This indicates that the model has a reasonable assessment of what it knows and what it doesn't know. Table I and Figure 4 show that the BNN-Auto method significantly reduces the standard deviation of the predicted logarithmic half-life from 0.418 to 0.280. This overall reduction in dispersion is a comprehensive manifestation of the improvement in the model's predictive ability and the more concentrated uncertainty estimation. A model that can systematically and significantly improve the prediction accuracy is more likely to provide uncertainty estimation that matches the distribution of prediction errors. As we clarified in the reply to Comment 12, factor 1.91 ($10^{0.280}$) stems from the standard deviation of the prediction residuals. Its statistical essence is that in the linear scale, the model's predicted values are highly likely to cluster within 1.91 times the range of the experimental value with a high probability. This quantitative description reflects the concentration degree of the model's prediction from another perspective, which is consistent with the accuracy target of uncertainty estimation.

7. Physical constraints on P_α: The preformation factor P_α is a probability and must lie within [0, 1]. However, the network architecture uses an ELU activation in the prediction head, which does not enforce this physical constraint. Do any predictions violate this bound? Are there values in Table II or III where $P_\alpha > 1$ or $P_\alpha < 0$? The authors should discuss how physical constraints are enforced or verified.

Response: Thank you for raising this important and detailed question. You have correctly pointed out that, as a probability, P_α should fall within the range of [0, 1], and you are concerned about whether the ELU activation function used for predicting the head can ensure this physical constraint. Firstly, we will provide an explanation from the perspective of model design. In our Bayesian neural network (BNN-Auto) framework, the final P_α prediction value is not directly obtained from the ELU activation output of a single forward propagation. As stated on page 8 of the original manuscript, during the prediction stage, we conducted 20 random forward propagations (Monte Carlo sampling), and the final prediction value P_α is the mean of these 20 sampling outputs ($P_\alpha = \sum_{t=1}^{20} H(x^*;\omega^{(t)})/20$). This integration prediction process itself has certain regularization and smoothing effects. More importantly, the training objective of our model ($P_\alpha^{\rm exp}$) is derived entirely from the experimental data via equation (6), and its values are strictly within the (0, 1) interval. During training, the model minimizes the error between the predicted values and the true values located within the (0, 1) interval (such as the negative log-likelihood), essentially learning to map the output to the vicinity of this interval. Although the ELU function itself does not limit the output, the entire Bayesian training framework (including the likelihood function based on physical data and variational inference) drives the model's prediction distribution to converge to the correct physical range. Secondly, and most importantly, we have carefully checked all the predicted results, including Table II and Table III. We can confirm that in all 535 nuclides listed and not listed in the tables, no $P_\alpha^{\rm BNN-Auto}$ prediction value has $P_\alpha > 1$ or $P_\alpha < 0$. All values strictly fall within the physically reasonable interval of 0 to 1. This empirically and strongly demonstrates that our BNN-Auto framework, despite employing nonlinear activation functions such as ELU in intermediate layers, produces final ensemble predictions that spontaneously and robustly satisfy the core physical constraint.

8. P_α from Eq. (6) can exceed unity: Equation (6) defines the experimental preformation factor as $P_\alpha^{exp} = T_{1/2}^{cal}/T_{1/2}^{exp}$. If the theoretical calculation overestimates the half-life relative to experiment, $P_\alpha^{exp} > 1$, which is physically problematic. How do the authors handle such cases? Are there nuclei in the dataset where this occurs, and if so, how are they treated?

Response: Thank you very much for your important insight. We fully understand and value your concern about data consistency. When inferring P_α^{exp} from experimental data, we did find that the P_α^{exp} of five nuclides was slightly greater than 1 in all 535 nuclei and the predicted data in Table III of the original manuscript, such as $^{158}Hf(P_\alpha^{exp} = 1.09)$,$^{226}Th(1.13)$,$^{230}Th(1.15)$,$^{232}U(1.03)$,$^{236}U(1.16)$. This phenomenon is not a data error but reflects the limitations of the ground-state theoretical model in certain specific nuclides. $T_{1/2}^{cal} > T_{1/2}^{exp}$ indicates that if strictly adhering to the assumption of $P_\alpha = 1$, the theoretical model underestimates the α decay rate of these nuclei. This is usually attributed to a slight overestimation in the calculation of the barrier penetration probability P (dependent on Q_α

and barrier shape) by the theoretical model, or the failure to fully capture certain special nuclear structure effects (such as shell effects, deformation) that favor the formation of α particles, thus leading to a longer calculated theoretical half-life. In our work, we did not simply discard the data with $P_\alpha^{exp} > 1$. These individual cases (accounting for less than 1% of the dataset) can be regarded as "indicators" of minor systematic deviations in the adopted fundamental theoretical model (CPT model) in certain nuclear regions. Keeping them intact in the training set helps the Bayesian neural network-autoencoder (BNN-Auto) framework we constructed to learn and correct this systematic deviation pattern from the data as a whole. The task of the BNN-Auto model is not to mechanically reproduce the P_α^{exp} values, but to learn the complex mapping relationship from nuclear features (Z, N, Q_α, l, etc.) to more reasonable preformation factors, which is essentially constrained by the physical common sense that $0 \leq P_\alpha \leq 1$. Although the labels P_α^{exp} of the input data slightly exceeded these 5 values by a little, the BNN-Auto model, through its loss function (based on likelihood) and by learning from a large amount of data $P_\alpha^{exp} \in (0, 1)$, naturally tended to regress to a reasonable physical range. More importantly, based on the previous question, we checked all the predicted results and confirmed that the model prediction values $P_\alpha^{\mathrm{BNN-Auto}}$ of all nuclides (including these 5) were strictly within the range of 0 to 1. This proves that our BNN-Auto framework can effectively integrate global data information, spontaneously produce physically consistent predictions, and correct the abnormal outputs of the basic theoretical model at individual points. These 5 samples account for a very small proportion, and the slight abnormality of their label values (the maximum being 1.16) had little impact on the overall learned patterns during the model's training with probability distribution weights. The main evaluation indicators of our model performance (such as the improved RMS in Table I) and the revealed physical laws (such as even-odd staggering, shell effect, etc.) have statistical robustness. They will not be affected substantially by this.

9. Prediction for $Z = 120$ nuclei: The predictions for $Z = 120$ superheavy nuclei (Table III) rely entirely on Q_α values from the WS4+ mass model. However, the uncertainties in these Q_α values propagate into the half-life predictions. The authors should discuss the sensitivity of their predictions to uncertainties in Q_α and consider using alternative mass models (e.g., FRDM, HFB) for comparison. Without this, the claimed verification of the $N = 184$ shell effect is premature.

Response: Thank you for your insightful suggestion. You correctly pointed out that the calculated decay energy Q_α value based on the quality model has inherent uncertainty, and this uncertainty will inevitably propagate to the final prediction of the half-life. This is crucial for reliable extrapolation predictions in the superheavy nucleus region. We fully agree that the sensitivity analysis of the uncertainty of Q_α and the comparison using different quality models can enable a more comprehensive assessment of the robustness of the prediction results. This is indeed a key direction that we plan to explore in our next work. Regarding this work, we would like to clarify a few points: Firstly, we chose the WS4+ quality model to provide the Q_α prediction values for the $Z = 120$ nuclide because it is widely considered to have high reliability in describing the mass of heavy and superheavy nuclei. It is a commonly used and mature benchmark model in related research fields (such as [N. Wang, M. Liu, X. Wu, and J. Meng, Phys. Lett. B 734, 215 (2014)] and its applications). Choosing a recognized and self-consistent quality model as the input is to establish a clear and repeat-

able prediction benchmark in our research. Secondly, the core objective of this work is to develop and validate the BNN-Auto framework in describing and predicting the α-particle preformation factor P_α. As shown in Table III and Figure 8 of the original manuscript, the key output of our prediction is $P_\alpha^{\mathrm{BNN-Auto}}$ and the resulting half-life $\log_{10}\mathrm{T}_{1/2}^{\mathrm{Pre}}$. To evaluate the performance of the P_α model itself on the same basis, we selected to compare our results with the prediction results of reference models (such as the ISM model) under the same Q_α input (all from the WS4+ model). This "controlled variable" comparison can more purely reveal the impact of different P_α processing methods on the half-life prediction. As shown in Figure 8, even under the same Q_α trend, the half-life curves given by BNN-Auto and ISM models both show a similar upward trend near $N = 184$, providing preliminary and mutually corroborating evidence for "this trend mainly originates from the shell physics effect reflected by the P_α model, rather than being dominated solely by the input Q_α trend". Finally, we sincerely thank you for your suggestion, which has pointed out a clear subsequent research direction for us. Systematically studying the uncertainty of Q_α (for example, through error propagation analysis or using the Monte Carlo method) and comparing the prediction differences brought by different quality models (such as FRDM, HFB series, etc.) will greatly enhance the credibility and physical depth of the superheavy nucleus prediction results.

10. Typo "half-half predictions" (p. 4): The sentence reads "improve the reliability of half-half predictions" — this should be "half-life predictions." While seemingly a typo, it appears in a key methodological sentence in the Introduction and reflects the overall need for more careful proofreading.

Response: Thank you for carefully reviewing our manuscript and pointing out this error. In the introduction section (page 3), the sentence "To address these limitations and improve the reliability of half-half predictions..." contains a typographical error. The correct expression should be "half-life predictions". We apologize for this oversight. We sincerely appreciate your suggestion that a more thorough review of the entire manuscript is needed. During the revision process, we conduct a systematic review and polishing of the entire manuscript to enhance its language quality and professionalism.

III. Minor Scientific Issues

11. The number of Monte Carlo samples ($T = 20$) used for posterior prediction (Eq. 9 and p. 9) seems low. The authors should justify this choice and demonstrate convergence of the predicted mean and variance with respect to the number of samples.

Response: Thank you for raising this important technical question. In the prediction stage of BNN-Auto, the selection of the number of Monte Carlo (MC) samples T used to approximate the posterior predictive distribution requires a balance between computational efficiency and statistical accuracy. We chose $T = 20$ based on the consideration of this balance point, and we believe that this choice is reasonable and sufficient for the prediction task of this study. First, the basis for choosing $T = 20$. In Bayesian deep learning practice, the MC sampling number T for posterior prediction is not always the larger the better. Its selection needs to consider two factors: one is the accuracy of statistical estimation and the computational cost, as described on page 9 and Equation (9) of the manuscript. We sample

the network weights $\omega^{(t)}$ from the variational posterior distribution $q(\omega|\theta)$ for T times and calculate the mean and standard deviation of the output $H(\mathbf{x}^*, \omega^{(t)})$ as the prediction value and uncertainty estimation. Theoretically, as T approaches infinity, the posterior prediction mean and variance can be obtained precisely, but the computational cost grows linearly. For the regression task (predicting P_α) involved in this work, our sampling number $T = 20$ can provide stable prediction statistics within an acceptable computational cost (the time for predicting 535 nuclides in the full dataset can be ignored). More importantly, the uncertainty of BNN prediction mainly comes from the weight distribution $q(\omega|\theta)$ itself, which characterizes the cognitive uncertainty. When the variational inference training converges, $q(\omega|\theta)$ has approached the true posterior, and at this time, extracting 20 independent and identically distributed samples from this distribution, their empirical mean and variance can well approximate the expected and variance of the theoretical distribution. The increase in T leading to an improvement in accuracy (reduction in MC error) has little impact on the macroscopic physical conclusions of this study. Secondly, supplementary verification of the convergence of the prediction mean and variance. We understand and agree with your suggestion that presenting the convergence behavior of the prediction statistics with increasing T empirically can further enhance the rigor of the methodology. Therefore, we have conducted supplementary analysis on the original manuscript in response to your question. We randomly selected multiple representative nuclides from the training set and validation set (covering even-even, odd-A, odd-odd types), systematically increasing the sampling number T from 5 to 200. For each T, we calculate the predicted standard deviation (σ) of P_α, and observe their trajectories as T changes. As shown in Figure 7 in the modification, when $T \geq 15$, the variation range of σ has dropped to a very low level (for example, when T increases from 20 to 100, the relative change in the predicted mean is typically less than 0.5%, and the relative change in the standard deviation is less than 2%). From $T = 20$ to larger values (such as 50 or 100), the prediction results do not show any significant deviation. This proves that for the BNN-Auto model in this study, the predicted mean and uncertainty estimation obtained using 20 samples have reached the actual convergence state. Increasing the sampling number does not change the physical prediction conclusion of the model, but only brings unnecessary computational cost.

12. The statement that predictions fall within "1.91 times the experimental error margin" (p. 13) appears without derivation or justification. How is this factor obtained? This needs clarification.

Response: Thank you for pointing out this statement. On page 13 of the manuscript, the statement "the vast majority of predicted values now fall within 1.91 times the experimental error margin" indeed requires a clear explanation of its derivation process to enhance the rigor of our work. As shown in page 13 and Figure 4 of the manuscript, using our BNN-Auto method, the standard deviation of the logarithmic half-life residuals (i.e., $\log_{10}\mathrm{T}_{1/2}^{\mathrm{BNN-Auto}} - \log_{10}\mathrm{T}_{1/2}^{\mathrm{exp}}$) has significantly decreased from 0.418 of the traditional CPT-LSM method to 0.280. To visually express this prediction accuracy on a linear scale, we calculated $10^{0.280} \approx 1.91$. This means that the majority of the model's calculated values are concentrated within approximately 1.91 times the experimental value on the linear scale. The expression 'experimental error margin' in the original text is not precise and may confuse. It actually refers to the statistical dispersion range of the BNN-Auto calculation values around

the experimental value. In the revised version, we have corrected it to a more accurate expression.

13. Figure 1 description: The flowchart, while informative, lacks specifics. The number of neurons in each layer, the dimensionality of the latent space, and the precise architecture details should be provided either in the figure or in a dedicated table.

Response: Thank you very much for your suggestions. Regarding the insufficient description of the specific configuration of each layer in the network of Figure 1 in the manuscript, the structural details of the autoencoder branch, and the coupling method between the Bayesian prediction module and the autoencoder, we have provided more detailed structural explanations in the revised version. In this paper, 8 input features are used as the network input. On this basis, the model first conducts representation learning through an autoencoder branch. The encoder part consists of three fully connected layers with the structure 8→128→64→8, that is, the 8-dimensional physical features are first mapped to a 128-dimensional hidden layer, then to a 64-dimensional hidden layer, and finally projected to an 8-dimensional latent representation. The corresponding decoder adopts a symmetrical structure 8→64→128→8, reconstructing the latent representation back to the original input space. Batch Normalization is equipped in the hidden layers of both the encoder and the decoder, and the ELU activation function is used to enhance training stability and nonlinear representation ability. In the prediction stage, the 8-dimensional latent representation output by the encoder is not used alone for regression, but is concatenated with the 8-dimensional reconstructed features output by the decoder to form a 16-dimensional fused representation, which is then sent to the subsequent shared prediction module. That is to say, in this version, the input of the shared prediction part consists of two parts: one part is the 8-dimensional latent variable extracted from the original physical input by the autoencoder, and the other part is the 8-dimensional reconstructed feature restored by the decoder. The purpose of this design is, on the one hand, to utilize the latent representation to provide high-level representation information after encoding and reconstruction constraints, and on the other hand, to utilize the reconstructed features to retain the overall structural information of the input variables, thereby enabling the prediction branch to complete the regression task under the joint effect of abstract representations and reconstruction information. Subsequently, this 16-dimensional fused representation enters the shared mapping module, which has the structure 16→128→64, and Batch Normalization and ELU activation functions are used in the hidden layers to further extract common feature representations suitable for subsequent regression. On this basis, we divide the samples into three categories according to the parity of the nucleons, namely even-even, odd-A, and odd-odd, and set up three parallel Bayesian expert prediction heads for these three types of samples. The structure of each expert head is 64→40→20→1, used to output the predicted value of $\log_{10}\mathrm{P}_{\alpha}$ for the corresponding sample. In other words, in this version, the category information is no longer explicitly concatenated with the latent representation through a low-dimensional embedding vector and sent into the shared module, but is reflected in the branch selection of the subsequent expert heads, that is, after the shared feature extraction, different types of nuclei are predicted by the corresponding expert sub-networks. Through this approach, the model retains the role of the autoencoder branch in structuring the representation learning of input features, as well as the ability to model the differences in prediction behaviors among different nuclear cat-

egories. It should be noted that the Bayesian modeling in this model is mainly reflected in the subsequent expert prediction heads, thereby providing corresponding uncertainty estimates while completing the regression prediction. Therefore, in the current BNN-Auto model, the role of the Autoencoder branch is mainly reflected in two aspects: first, through the reconstruction process of the encoder 8→128→64→8 and the decoder 8→64→128→8, it imposes representation constraints on the input features, thereby learning more stable and structurally constrained latent representations; second, by sending the latent representation and the reconstructed features together into the shared prediction module, the network can simultaneously utilize abstract representation information and input reconstruction information. Subsequently, by combining the Bayesian expert predictions corresponding to the three types of nuclides, the final prediction of $\log_{10}\mathrm{P}_\alpha$ and its uncertainty modeling are achieved.

14. The claim of a "possible neutron subshell closure" at $N = 152$ (p. 16): This is a well-known phenomenon in nuclear physics already established experimentally and theoretically. The phrasing "possible" is misleading. The authors should properly reference the existing literature on the $N = 152$ deformed shell closure.

Response: Thank you very much for your important and professional comment. You correctly pointed out that the shell (or subshell) effect at neutron number $N = 152$ in the region of heavy and superheavy nuclei is a well-established physical phenomenon that has been extensively verified through numerous experiments and theoretical studies. The use of the word "possible" in the original manuscript on page 16 to describe this effect was indeed not precise enough. It might have misled readers into thinking that the phenomenon was still controversial or not yet confirmed. In the revised manuscript, we have removed the word "possible" from the phrase "possible neutron subshell closure" and directly stated the phenomenon. We have also added a reference to the latest theoretical work specifically studying this phenomenon [S. Tang, Y. Qian, T. Wan, and Z. Ren, Phys. Rev. C 112, 014308 (2025).] at the corresponding discussion point.

15. The paper lacks error bars on the $P_\alpha^{BNN-Auto}$ values in Tables II and III. Since the BNN framework naturally provides uncertainties, these should be reported alongside the mean predictions.

Response: Thank you for raising this important point. We fully agree that the predictive uncertainty information provided by the Bayesian neural network framework is one of its core advantages and should be clearly presented in the results. In the revised manuscript, we have supplemented the uncertainty for $P_\alpha^{BNN-Auto}$ in Tables II and III, directly presenting the complete prediction values including asymmetric uncertainty obtained through posterior distribution sampling. We believe that the information currently provided in this column (the central estimate and its upper and lower bounds) is sufficient to meet the needs of evaluating the reliability of predictions. Adding a separate column for "average value" might cause information redundancy, and this central estimate is already implicitly included in this representation. The current presentation of the tables in the revised manuscript can clearly and completely convey the posterior information of Bayesian predictions, including their central tendency and dispersion. Thank you again for your detailed review, which has helped us enhance the rigor and completeness of the result presentation.

IV. English Language and Grammar Issues

The manuscript contains numerous grammatical errors, awkward constructions, and unclear sentences that substantially hinder readability. I list representative examples below, but a thorough proofreading by a native English speaker or a professional editing service is strongly recommended before resubmission.

A. Grammatical Errors

16. p. 3: "the half-lives of α decay is commonly expressed" $\rightarrow$ "the half-life of α decay is commonly expressed" or "the half-lives ... are commonly expressed" (subject-verb agreement).
17. p. 3: "can treated with" $\rightarrow$ "can be treated with" (missing auxiliary verb).
18. p. 3: "P_α, involves the degree of α-clustering...is a complex quantum many-body problem" — the comma after P_α creates a sentence fragment. This should be restructured: "P_α involves the degree of α-clustering...and is a complex...".
19. p. 4: "improve the reliability of half-half predictions" $\rightarrow$ "half-life predictions".
20. p. 7 (Section III): "In this work, it is achieved a high-precision description" $\rightarrow$ "In this work, a high precision description is achieved" or "This work achieves a high-precision description".
21. p. 8: "Input features comprise physical quantities such as proton number N, neutron number Z" -the standard convention is Z = proton number and N = neutron number. This appears to be a transposition error.
22. p. 9: "there are still exhibit certain deviations" $\rightarrow$ "there still exist certain deviations" or "they still exhibit certain deviations".
23. p. 12: "the calibrate and original calculations" $\rightarrow$ "the calibrated and original calculations".
24. p. 18 (Summary): "In this work, a novel framework for describing the P_α is constructed by integrating the BNN-Auto"- the phrasing "the BNN-Auto" without specifying "method" or "framework" is awkward.

Response: Reply to your regarding grammatical and stylistic corrections (Points 16-24):
We sincerely thank you for the meticulous reading and for identifying the grammatical errors, typos, and awkward phrasings throughout the manuscript. We have carefully revised the manuscript and have corrected all the specific points raised:
p. 3: "the half-lives of α decay is..." has been corrected to "α decay half-lives are...".
p. 3: "can treated with" has been corrected to "can be treated with".
p. 3: The sentence fragment regarding P_α has been restructured for clarity.
p. 4: The type "half-half predictions" has been corrected to "half-life predictions".
p. 7 (Section III): The awkward construction "it is achieved a..." has been rephrased to "a high-precision description is achieved".
p. 8: The transposition error in "proton number N, neutron number Z" has been corrected to the standard convention: "proton number Z, neutron number N".
p. 9: "there are still exhibit" has been corrected to "they still exhibit".
p. 12: "the calibrate and original calculations" has been corrected to "the calibrated and original calculations".
p. 18 (Summary): The phrasing "integrating the BNN-Auto" has been clarified to "inte-

grating the BNN-Auto framework".

These corrections significantly improve the clarity, readability, and professionalism of the manuscript. We are grateful for the reviewer's attention to detail, which has helped us polish the final manuscript.

B. Sentence Construction and Style

25. Throughout the paper, sentences are often excessively long and convoluted. For example, the sentence beginning "To address these limitations and improve..." on p. 4 runs for 6 lines and tries to convey too much information at once. Such sentences should be split for clarity.
26. Inconsistent use of articles: "the BNN-Auto method" vs. "BNN-Auto" without article; "the half-life" vs. "half-life" used inconsistently. The authors should settle on consistent usage.
27. Redundant phrasing: Phrases like "in this processing" (p. 3, which should be "in this process"), "intuitively proves that" (p. 11), and "clearly reveals inherent systematic biases" overuse emphatic adverbs. Scientific prose benefits from more restrained language.
28. The Abstract uses "α-particle preformation (P_α)" where P_α is actually the preformation factor or probability, not the preformation itself. This shorthand is potentially confusing and should be clarified.

Response: Reply to your regarding sentence construction and style (Points 25-28):
We greatly appreciate your insightful comments on improving the sentence structure, style, and clarity of the manuscript. We have thoroughly revised the manuscript to address each of the points raised:
25. Long and convoluted sentences: We have carefully reviewed the manuscript and split overly long sentences (including the example noted on p. 4) into shorter, clearer statements to enhance readability and comprehension.
26. Inconsistent use of articles: We have standardized the terminology throughout the manuscript. We now consistently use "the BNN-Auto method/framework". Similarly, the usage of "the half-life" vs. "half-life" has been made consistent based on context.
27. Redundant phrasing and overuse of emphatic adverbs: We have revised the text to adopt more restrained and precise scientific language. Phrases like "in this processing" have been corrected to "in this process", and emphatic adverbs such as "intuitively", "clearly", etc., have been removed or replaced where they did not add substantive meaning, resulting in a more direct and professional tone.
28. Potentially confusing shorthand in the Abstract: The phrasing in the Abstract has been clarified. The shorthand "P_α" is now explicitly introduced as "the α-particle preformation probability (P_α)" upon its first use, to avoid any potential confusion.

These revisions have significantly improved the flow, consistency, and overall quality of the writing. Thank you for these valuable suggestions, which have helped us polish the manuscript to a higher standard.

V. Presentation, Figures, and Tables

29. Figures 5–7: These 3D surface plots are visually cluttered and difficult to read. The axis labels are small and partially obscured. Standard 2D line plots with separate panels for P_α and half-lives would be far more informative. If the authors insist on 3D plots, they should increase label font sizes and ensure all data points are visible.

Response: Thank you for your valuable suggestions. The 3D diagrams in the original manuscript were visually rather chaotic and did not facilitate the clear presentation of the data results. To significantly enhance the readability of the diagrams and the effectiveness of information transmission, we have strictly followed your suggestions and replaced all the Figures 5-7 in the original manuscript with standard 2D diagrams (the modified Figures 8-10) in the revised version.

30. Figure 3: The nuclide chart comparison is a strong visual, but the color scale saturates too quickly. Consider using a diverging colormap with a wider range to better resolve the residual structure.

Response: Thank you for this valuable suggestion. We fully agree that optimizing the color scale can significantly enhance the resolution and visual clarity of the residual structure in the nuclide diagram. According to your suggestion, we have made corresponding improvements to Figure 3 in the revised manuscript: We have used a blue-white-red divergent color mapping, centered around the zero value (white), to distinguish positive and negative residuals visually. We have redefined the data range of the color bar. The new range is determined based on the statistical distribution of the residual data, ensuring that the color gradient can be better utilized and effectively avoiding the problem of premature color saturation. This enables the residual systematic deviation patterns remaining in the nuclear shells (such as near N=126, Z=82) and the uniform distribution corrected by the BNN-Auto method to be clearly presented with more delicate color gradations. These modifications significantly enhance the readability and information content of Figure 3, allowing us to present the important conclusion of our work, namely that the BNN-Auto method can effectively capture and correct the systematic deviations related to traditional nuclear structure effects, more intuitively and powerfully. Once again, thank you for helping us improve the quality of the manuscript figures.

31. Figure 8: The predicted half-lives for $Z = 120$ should include uncertainty bands from the BNN framework, which would be one of the key added values of the Bayesian approach.

Response: Thank you for your valuable suggestions. As you pointed out, using the predictive uncertainty provided by the Bayesian neural network framework to draw uncertainty bands is a key advantage of this method over traditional deterministic models. It enables a more intuitive assessment of the reliability of the prediction results in unknown regions. In the revised manuscript, we have followed your advice and added the corresponding uncertainty bands to the prediction results of BNN-Auto in Figure 11 (originally Figure 8 in the original manuscript), which shows the predicted half-lives of α decay for superheavy nuclei with $Z = 120$. This addition makes our figure more complete in presenting the output of the BNN-Auto model, providing not only point predictions but also the confidence intervals for each prediction. This is particularly important for evaluating the reliability of theoretical predictions in the superheavy nucleus region.

32. Table II: The table spans two pages and is split across a page break. It should be reformatted for continuity or clearly labeled as "Table II (continued)" which is done, but the formatting could be improved with repeated column headers.

Response: Thank you for your meticulous review and suggestions. We fully agree that repeating the complete column titles in the continuation pages of the cross-page table can significantly enhance the readability of the table. Based on your suggestion, we have optimized the formats of Table II and Table III in the revised version.

33. Table III: The ISM reference values are listed but the ISM method is never formally introduced or defined in the text. The abbreviation "ISM" appears only in Table III and Figure 8 without being spelled out.

Response: Thank you for pointing out this omission in detail. When using an abbreviation for the first time in the manuscript, its full definition should be provided. In the manuscript, "ISM" is the abbreviation for "Improved Simple Model". This model is a theoretical model we proposed in our previous work for calculating the half-life of α decay ([X.Y. Zhu, S. Luo, W. Gao et al., Chin. Phys. C 48, 074102 (2024)]). The model improves the harmonic oscillator potential model (HOPM) by introducing a centrifugal potential correction term, thereby being able to uniformly describe the half-lives of both favored and unfavored α decay. In this work, we refer to the ISM model as a reliable benchmark model for comparison and verification when predicting the half-life of superheavy nuclei ($Z = 120$) for α decay, in order to highlight the effectiveness and rationality of the BNN-Auto method. Based on your suggestion, we have provided supplementary explanations for ISM in the revised manuscript.

VI. References and Literature

34. Reference [63]: The Adam optimizer reference is cited as "K. D. B. J. Adam et al., arXiv preprint arXiv:1412.6980 1412 (2014)". This is incorrectly formatted. The correct reference is: D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," arXiv:1412.6980 (2014). The current citation is garbled and unprofessional.
35. Reference [64]: This reference is given only as a DOI (10.1016/j.inffus.2021.11.005) without author names, title, journal, volume, or page numbers. This is incomplete and violates PRC reference formatting standards.
36. Missing key references: The paper does not cite several important works on machine learning applications in nuclear physics, including: (a) the seminal work by Mumpower et al. on BNN for nuclear masses, (b) recent work by Lovell et al. on Bayesian approaches to nuclear reactions, and (c) the broader review by Boehnlein et al. (Rev. Mod. Phys. 94, 031003, 2022) on ML in nuclear physics. Given the PRC editor's explicit request to evaluate whether the authors make full use of the available literature, this is a notable deficiency.
37. Self-citation concentration: References [38], [51], [69] are all from overlapping author groups. While self-citation is acceptable when relevant, the paper should also engage more thoroughly with competing approaches from other groups working on α-decay predictions with ML methods.
38. The $N = 152$ deformed shell closure is well documented in the literature (e.g., Herzberg et al., Nature 2006; Heenen et al.). The authors should cite these foundational

works when discussing their observation near $N = 152$.

Response: Thank you for carefully reviewing the references and providing valuable suggestions. We have, based on your suggestions, conducted a comprehensive review and supplementation of the reference section in the revised manuscript. The specific modifications and explanations are as follows:
34. Reference [63] and 35. Reference [63] (reference format issue): We fully agree with your opinion. Inappropriate citation formats can affect the rigor of the paper. In the revised manuscript, we have corrected these two citations, namely corresponding to [66] [D. Kinga, J. B. Adam, et al., in International conference on learning representations (ICLR), Vol. 5 (California, 2015).] and [67]: [Y. Tian and Y. Zhang, Inform. Fusion 80, 146 (2022).]
36. (Missing Key Literature). Thank you for pointing out this significant oversight. We agree that extensively citing landmark and review works within the field is necessary to demonstrate the completeness of the research background. In the revised introduction section of the manuscript, we have supplemented the citation of the key literature you suggested: [52][A. Boehnlein, M. Diefenthaler, N. Sato, M. Schram, V. Ziegler, C. Fanelli, M. Hjorth-Jensen, T. Horn, M. P. Kuchera, D. Lee, et al., Rev. Mod, Phys. 94, 031003 (2022).]; [53][M. Li, M. Mumpower, N. Vassh, W. S. Porter, and R. Surman, Phys. Rev. Res. 7, L032057 (2025).]; [54][G. B. King, A. E. Lovell, L. Neufcourt, and F. M. Nunes, Phys. Rev. Lett. 122, 232502 (2019).].
37. (Self-citation Concentration): We understand and appreciate your attention to this issue. We refer to references [38], [51], [72] (i.e., [69] of the previous draft) because one of the core goals of this work is to systematically introduce and evaluate the performance improvement brought by the Bayesian neural network and autoencoder (BNN-Auto) method (see reference [38]). Therefore, the direct comparison with the CPT-LSM model is the most direct and necessary basis for evaluating the effectiveness of the method proposed in this work. This led to a relatively concentrated situation of self-citation in the method comparison section. At the same time, we fully agree that an excellent paper should fully demonstrate the connection and comparison with other mainstream methods in the field. To this end, in the revised draft, the discussion section (such as the comparison with other machine learning methods in Section III) has been supplemented with references to more works by other teams using machine learning to study alpha decay or nuclear structure problems, to reflect the breadth of the research background better. We will systematically apply this method to more different theoretical models in future research and conduct systematic comparisons with a wider range of international peers to verify further and expand the universality of this method.
38. (Reference for N=152 shell closure): Your point is entirely correct. When discussing the possible neutron subshell closure effect near $N = 152$, it is necessary to cite the pioneering experimental and theoretical work recognized in the field. We have added the citation at the corresponding position in the revised manuscript: [83] [R. D. Herzberg, P. T. Greenlees, P. A. Butler, et al., Nature 442, 896 (2006).]; [84] [S. Cwiok, J. Dobaczewski, P. H. Heenen, P. Magierski, and W. Nazarewicz, Nucl. Phys. A 611, 211 (1996).].
Once again, we would like to express our gratitude for your meticulous and constructive suggestions. These recommendations have greatly helped us improve the standardization, completeness, and academic rigor of the references and significantly enhanced the quality of the paper.

VII. Logical Coherence and Flow

39. The Introduction (Section I) is well-structured in its first half but becomes disjointed when transitioning from the discussion of traditional models to the ML approach. The paragraph beginning "Among various machine learning methodologies" on p. 4 abruptly shifts focus without a smooth transition.

Response: Thank you very much for your meticulous review and valuable suggestions. We fully agree with your opinion. The original text does indeed have a gap in the transition when shifting from the traditional model discussion to the machine learning approach. To make the logic more coherent, in the revision of the manuscript, after the sentence "Thus, there is an urgent need to establish a novel method capable of adaptively learning the systematic trends of nuclear structure from data, in order to globally describe P_α and accurately predict the α decay half-lives of superheavy nuclei.", we added the following transitional sentence: "To address this challenge, machine learning methods that can adaptively learn from data and quantify the uncertainty of predictions, especially Bayesian neural networks, offer a highly promising solution." This modification clarifies the purpose of introducing machine learning methods, which is to directly address the limitations of the traditional model proposed in the previous text, thus making the logical transition between the paragraphs smoother and more natural.

40. There is a logical gap between the description of the BNN-Auto architecture (Section II.B) and its application (Section III). The feature engineering choices (the three derived features λ_1, λ_2, λ_3) are mentioned but not justified physically. Why these specific combinations? What is the physical motivation for $Z/\sqrt{Q_\alpha}$? This connection between ML methodology and nuclear physics should be made explicit.

Response: Thank you very much for your detailed and insightful review comments. The issue you pointed out regarding the physical motivation for the features (particularly the three derived features $\lambda_1, \lambda_2, \lambda_3$) is crucial for establishing a solid bridge between machine learning methods and nuclear physics. We fully agree and have provided supplementary explanations in the corresponding section of the revised manuscript (Section II.B "Bayesian neural network with autoencoder"). In the α decay theory, the preformation factor P_α is influenced by both the nuclear structure (such as shell effects, pairing correlations) and the decay dynamics. The 8-dimensional input features we chose are designed to encode this core physical information. Among them, the three basic features (A, N, Z) directly describe the global properties of the parent nucleus. The three derived features $(\lambda_1, \lambda_2, \lambda_3)$ are constructed based on the profound considerations of α decay physics. In the droplet model and many nuclear potentials, the radius of the nucleus is proportional to λ_1. This feature converts the mass number into a scaling related to the nuclear size, enabling the network to more directly learn the influence of the geometric scale of the nucleus on the preformation probability of α particles. λ_2, this dimensionless quantity, intuitively characterizes the relative strength ratio of Coulomb repulsion and decay kinetic energy. It enables the model to more efficiently capture how these two competing factors jointly determine the decay rate and thereby affect the P_α value obtained from the half-life. λ_3 is the scaling form of the eigenvalue of the angular momentum operator $(\hbar l(l+1))$ in quantum mechanics, which is the physical manifestation of angular momentum. In potential barrier calculations, it can more accurately reflect the

contribution of the centrifugal barrier to the effective potential than l itself, enabling the model to handle non-zero angular momentum decays more precisely.

41. The paper claims to "verify the shell effect of the predicted stable island" (Abstract), but the prediction relies on Q_α from WS4+ which already encodes shell effects. The BNN-Auto is then applied on top. The logic is somewhat circular: the shell structure input to Q_α drives the shell structure output in half lives. The authors should address this potential circularity more carefully.

Response: Thank you for your profound and insightful suggestions. The issue you pointed out regarding "logical circularity" is extremely crucial, prompting us to examine and clarify the expression of our conclusion more rigorously. In the revised manuscript, we will modify the abstract and the relevant expressions in Section III, changing "verifies the shell effect" to a more accurate description. This modification clarifies that the WS4+ input already includes the shell effect, and the contribution of this work lies in proving that the BNN-Auto framework can provide self-consistent and reliable extrapolation predictions based on such physical input, thereby avoiding logical ambiguity. Once again, thank you for your valuable suggestions.

42. Section III discussion of Figs. 5–7: The analysis of odd-even staggering and shell effects, while physically reasonable, does not sufficiently distinguish between what the model has learned from the training data versus what constitutes a genuine prediction. For nuclei within the training set, agreement with experimental trends is expected. The paper would benefit from clearly separating in-sample performance from out-of-sample predictions.

Response: Thank you for your valuable suggestions. The point you made about clearly distinguishing "in-sample performance" from "out-of-sample prediction" is absolutely correct. This will make the logical structure of the paper more rigorous. The analysis in Figure 8-10 of the revised manuscript aims to demonstrate that the BNN-Auto model can successfully learn and capture the known nuclear structure patterns (such as parity effect and shell effect) from the training data. This belongs to the in-sample validation of the model and is the basis for testing the physical interpretability of the model. The true out-of-sample prediction of this work is reflected in the extrapolation calculation of the half-life of the unknown superheavy nucleus ($Z = 120$) in the latter part of Section III. To clearly distinguish this, we have supplemented the discussion on Figure 8-10 in Section III of the revised manuscript. This supplementation will clearly differentiate that the former part is the verification of the model's capabilities and the latter part is the prediction for new regions, thereby avoiding possible confusion. This modification has made the logical structure of our work more distinct. Once again, we thank you for your meticulous review.

VIII. Additional Suggestions for Improvement

43. Provide the code and/or trained model: In the spirit of reproducibility, the authors should consider making their code and trained model weights publicly available, e.g., via a GitHub repository or supplementary material.

Response: Thank you for your valuable suggestions. We fully understand and agree with

the importance of open science and reproducible research, which is crucial for promoting the development of the field. We have made the complete code repository and trained model weights used in this work publicly available on GitHub (https://github.com/Lin-lab-code/BNN) to ensure that other researchers can fully reproduce our results and conduct further studies.

44. Include a table of hyperparameters: The network architecture details (number of layers, neurons per layer, latent dimension, activation functions, k value, learning rate schedule) should be collected in a single summary table for reproducibility.

Response: Thank you for your valuable suggestions. Clear explanations of hyperparameters are crucial for the reproducibility of the work. In our revised manuscript, we have, in accordance with your opinion, concentrated and clarified the core details of the network architecture and training hyperparameters in Section II. B. The encoder-decoder architecture (dimensions: $8 \to 128 \to 64 \to 8$, $8 \to 64 \to 128 \to 8$) and the corresponding activation functions (ReLU, Sigmoid, ELU), as shown in Figure 1. Optimizer parameters (AdamW, learning rate 10^{-4}, weight decay 10^{-6}, batch size 32) and the number of training epochs (1000 epochs). The weight factor of the loss function is $k = 0.2$ (its sensitivity analysis is shown in Figure 6). This ensures that all key information is easily accessible, fully meeting the requirements for reproducibility, and also avoids the potential text redundancy caused by adding independent tables. Once again, thank you for helping us further improve the clarity of the manuscript.

45. Discuss computational cost: How long does training take? How does this compare to the CPT-LSM approach? What hardware was used? This context is important for readers who may wish to apply similar methods.

Response: Thank you for your meticulous review and the valuable questions you raised. The BNN-Auto model of this work is trained based on a dataset of 535 nuclides (with 8 input features). All deep learning experiments are conducted on a workstation equipped with an NVIDIA GeForce RTX 3090 (24GB) GPU, an Intel Xeon Platinum 8272CL CPU, and approximately 376GB of memory. For the final version of the BNN-Auto network in the revised manuscript, a complete 10-fold cross-validation training and validation are performed on a single RTX 3090, and the actual total time consumed is approximately 436.7 seconds (about 7.3 minutes), with an average training time per fold being approximately 43.6 ± 11.5 seconds. In contrast, the computational cost during the prediction stage is very low: a single Monte Carlo posterior prediction with $T = 20$ for all 535 nuclides only takes about 0.05 seconds. Regarding the comparison with CPT-LSM, it should be noted that the calculation paradigms of the two are different. CPT-LSM belongs to a parametric method based on physical models and least squares fitting, with its parameter scale much smaller than that of neural networks and not involving multiple rounds of backpropagation optimization. Therefore, the computational cost is significantly lower than that of BNN-Auto. In other words, CPT-LSM is more computationally lightweight. At the same time, BNN-Auto achieves stronger nonlinear fitting ability, uncertainty quantification ability, and better prediction performance at the cost of higher training cost.

46. Consider adding a SHAP or feature importance analysis: Since the paper aims

to extract physical insights from the ML model, an analysis of which input features most strongly influence the predictions would strengthen the interpretability claims.

Response: Thank you for your valuable suggestions. We fully agree with your viewpoint on enhancing the interpretability of the model, which is also one of the core goals of our work. In this paper, we mainly achieve the interpretability of the model through two closely integrated physical approaches, rather than relying on post hoc feature importance scores: Firstly, in Section II. B of the manuscript, we clearly expound the physical motivation of the input features. Particularly, the three derived features $(\lambda_1, \lambda_2, \lambda_3)$ were directly constructed based on the α decay theory (such as radius scaling, the competition between Coulomb energy and decay energy, and the physical form of angular momentum). This design ensures that the model learns from the very beginning in a feature space with clear physical meaning, and its "importance" has been pre-set and explained by physical theories. Secondly, in Section III, we systematically verified the consistency of the model's predictions with known nuclear physics laws. As shown in Figures 8-10 of the revised manuscript, the model successfully reproduced key nuclear structural laws such as parity effects and shell effects. This indicates that the model is not performing "black box" fitting but truly learning the systematic laws with clear physical meanings that we expect. This is more strongly able to prove the interpretability and physical credibility of the model than feature importance scores. The combination of "physically-driven feature design" and "reproduction verification of physical laws" has provided a solid and conforming to the nuclear physics research paradigm explanation for the reliability of the BNN-Auto model from both the mechanism and appearance levels. Adding relatively general feature importance analysis might deviate from the main discussion line of this paper, which is dominated by physical discovery and verification, and might make the focus less prominent. We once again thank you for your profound insights, which have prompted us to more clearly articulate the substantive work we have completed in terms of interpretability in this paper. We hope the above explanations can address your concerns.

47. Proofread mathematical notation: Some formulas in the PDF appear with missing symbols (e.g., the title mentions "-particle" instead of "α-particle" in the referral letter). Ensure all Greek letters render correctly in the final submission.

Response: Thank you very much for your meticulous review. We have thoroughly checked and corrected all the mathematical symbols, Greek letters, and formatting throughout the document based on your comments. We have ensured the consistency of the format for all physical quantities, variables, and mathematical expressions in the text.

IX. Summary of Required Revisions

Before this manuscript can be reconsidered for publication in Physical Review C, the authors must address the following critical items:

Essential (must address):

- Correct the N/Z notation inconsistency throughout the manuscript.
- Add ablation study: BNN without autoencoder vs. BNN-Auto vs. standard NN with

autoencoder.
• Report and analyze the hyperparameter k and all architecture details.
• Provide uncertainty calibration analysis for the BNN predictions.
• Include error bars / uncertainty bands in predictions (Tables II, III; Fig. 8).
• Fix all reference formatting issues (Refs. [63], [64]).
• Comprehensive English language revision.
• Address the circularity concern regarding the $Z = 120$ shell effect claim.

Strongly recommended:

• Compare with at least one other ML method (e.g., standard BNN, Gaussian process).
• Replace 3D surface plots (Figs. 5–7) with clearer 2D representations.
• Discuss sensitivity of $Z = 120$ predictions to the choice of mass model for Q_α.
• Add missing references to ML-in-nuclear-physics literature and $N = 152$ shell closure.
• Define ISM abbreviation and discuss it properly in the text.
• Address the physical constraint $P_\alpha \in [0, 1]$ and how it is enforced.

Response: Thank you very much for summarizing the revision requirements for Part IX. We have made comprehensive and meticulous revisions to the manuscript based on all your suggestions. The revised content has been integrated into the revised manuscript, and for each of the 'must address' and 'strongly recommended' requirements, we have detailed the modification measures and their specific positions in the text below.
Essential (must address):
(1) Corrected the inconsistent N/Z symbols throughout the text: We have thoroughly checked and corrected the issues of incorrect usage of proton number (Z) and neutron number (N) symbols in the entire text (particularly on page 9), ensuring that they comply with the nuclear physics standard specifications (where Z represents the proton number and N represents the neutron number).
(2) Supplementary ablation experiment analysis: In the "Results and Discussion" section, we have added a systematic comparison and ablation experiments. As shown in Figure 5, we compared BNN-Auto with various mainstream machine learning methods (random forest, Gaussian process, standard feedforward neural network, etc.), demonstrating the superiority of BNN-Auto. At the same time, we constructed a standard BNN (without autoencoder, S-BNN) and a model with disabled KL regularization (KL off) for ablation experiments, quantitatively evaluating the contributions of the autoencoder and the Bayesian framework to the performance improvement. (Pages 14-15 of the manuscript, Figure 5 and related discussions.)
(3) Report and analyze the hyperparameter k and all architectural details: On page 15, we clearly state that $k = 0.2$ was selected through grid search, and provide a detailed sensitivity analysis (Figure 6), which shows the stability of the model performance within the range of $k \in [0.1, 0.6]$, and discusses the key role of KL regularization therein; In the "B. Bayesian neural network with autoencoder" subsection on page 6, we describe the network architecture in a more focused manner: the activation functions used by the encoder, decoder, and prediction head, as well as the training details. (Manuscript pages 6, 15, Figures 1, 6.)
(4) Uncertainty calibration analysis of BNN predictions: We verified the stability of the

prediction uncertainty estimation through empirical convergence analysis (Figure 7). This figure indicates that when the Monte Carlo sampling time $T \geq 15$, the prediction mean and standard deviation (i.e., uncertainty) have basically converged. Using $T = 20$ is sufficient to obtain a stable statistical estimate. Additionally, the results in Figure 2, Figure 4, and Table I (the prediction intervals have significantly narrowed and covered the experimental values, and the standard deviation of the prediction residuals has decreased from 0.418 to 0.280) indirectly but strongly support the rationality of the uncertainty estimation. (Page 16 of the manuscript, Figure 7; and supplementary discussions on Figures 2, Figure 4, and Table I on pages 8, 11, and 13.)
(5) Add error bars/uncertainty intervals to the prediction results: We have supplemented the uncertainty information for all key prediction results. Table II and Table III: The column "$P_{\alpha}^{\mathrm{BNN-Auto}}$" now reports the posterior prediction uncertainty in the format of "$central value^{upper bound}_{lower bound}$". Figure 11 (original Figure 8): The prediction curve of BNN-Auto now includes an error band based on the prediction standard deviation, visually demonstrating the confidence interval of the prediction. (Descriptions of Figure 11 on pages 11-12 of the manuscript, pages 21-23 of Table III, and page 21 of Table II.)
(6) Correction of reference citation format: We have thoroughly checked and corrected the citation formats, particularly for the original Ref. [63] (now manuscript [66], Kingma and Ba, Adam optimizer) and the original Ref. [64] (revised manuscript [67], Tian and Zhang, 2022), to ensure they conform to the journal's standards.
(7) Comprehensive English language revision: We have meticulously refined the entire text, correcting the grammatical errors, spelling mistakes, lengthy sentence structures, and inconsistent word usage that you pointed out, as well as those we identified through our own review. For instance, we changed "half-half predictions" to "half-life predictions", smoothed out awkward sentence structures such as "it is achieved a...", and standardized the usage of terms (such as "the BNN-Auto method").
(8) Clarification of the logical circularity doubts regarding the statement about the $Z = 120$ shell effect: We fully agree with your concerns and have provided a more precise expression in the text. In the abstract and on page 21, we have changed "verifies the shell effect..." to "The consistency between the predicted half-life trend and the inherent shell-stabilization signature in the WS4+ inputs corroborates the predictive reliability of the BNN-Auto framework for the hypothesized "island of stability". This clearly indicates that our contribution lies in demonstrating that the BNN-Auto framework can provide self-consistent and reliable extrapolation predictions based on the physical input (WS4+ Q_{α}) that already includes the shell effect, rather than independently "discovering" the shell effect.

Strongly recommended:
(1) Compared with at least one other ML method: As stated in the second item of "Must Handle", we have systematically compared BNN-Auto with methods such as Random Forest (RF), Gaussian Process (GP), Standard Feedforward Neural Network (SFNN), and Linear Least Squares Regression (L-LSR) in Figure 5, demonstrating the superiority of BNN-Auto in terms of prediction accuracy. (Modified pages 14-15 of the manuscript, Figure 5.)
(2) Replace the 3D surface graph with a clearer 2D diagram: We have adopted your suggestion and replaced all the 3D surface graphs in the original manuscript (original figures 5-7)

with standard 2D line graphs (new figures 8-10), which has greatly improved the readability and information transmission efficiency. (Revise pages 17-20 of the manuscript, figures 8-10.)
(3) Discussing the sensitivity of the $Z = 120$ prediction to the selection of the quality model (for Q_α): In our reply letter (pages 6-7) and the main text discussion, we acknowledge that systematically studying the uncertainty of the Q_α input and its sensitivity to different quality models is an important direction for future work. In this current work, to establish clear and reproducible comparison benchmarks, we chose the widely recognized WS4+ model. We compared the effects of different P_α processing methods under the same Q_α input, in order to more purely evaluate the predictive ability of the BNN-Auo framework itself.
(4) Additional key references on the application of machine learning in nuclear physics and the closure of the $N = 152$ shell: We have supplemented the key references in the introduction and discussion sections. (The references [52], [53], [54], [83], [84] at the end of the text and the corresponding citations in the main text.)
(5) Define the abbreviation "ISM" and appropriately discuss it in the text: When "ISM" first appeared on page 21 of the revised manuscript, we had clearly stated its full name as "improved simple model" and provided a reference (Zhu et al., Chin. Phys. C 48, 074102 (2024)) to explain its role as a comparison benchmark model.
(6) Discuss the physical constraint $P_\alpha \in [0, 1]$ and how it is guaranteed: In our reply letter (page 5) and the argument in the main text, we explain that although the use of the ELU activation function in the prediction head does not inherently enforce the output range, the BNN-Auo framework trains through a likelihood function based on physical data ($P_\alpha \in (0, 1)$), driving the posterior prediction distribution to converge to the correct physical range. Empirically, we checked all the prediction results (including Table II and III) and confirmed that none of the kernel's $P_\alpha^{\mathrm{BNN-Auto}}$ values exceeded the range of [0,1]. The model spontaneously learned this constraint from the data.

We have carefully, and item-by-item, addressed all the revised requirements you summarized as 'must address' and 'strongly recommended'. These modifications have significantly enhanced the scientificity, rigor, and readability of the manuscript. Once again, we sincerely thank you for your insightful review and valuable guidance.

We have made modifications according to your comments, which are shown in blue color in the updated version. We hope our responses and modifications can clarify your concerns, and would like to thank you again for your precious time on promoting the quality of this work.

Sincerely yours,

Xiao-Yan Zhu on behalf of all the authors.